

ELEC 221 Tutorial 3 – Solutions

CT Impulse, Convolution Integral, Impulse Response Properties, and Frequency Response

Monday, 28 September 2026

1. CT impulse, unit step, and the convolution integral
2. The impulse response and system properties
3. Exponential signals and frequency response

$$1.1 - x(t) * \delta(t) = ?$$

Expected outcome: convolving with $\delta(t)$ leaves the signal unchanged – $\delta(t)$ is the identity element of convolution, just as 0 is the identity of addition.

$$1.1 - x(t) * \delta(t) = ?$$

Expected outcome: convolving with $\delta(t)$ leaves the signal unchanged – $\delta(t)$ is the identity element of convolution, just as 0 is the identity of addition.

$$x(t) * \delta(t) = \int_{-\infty}^{\infty} x(\tau) \delta(t - \tau) d\tau$$

1.1 – $x(t) * \delta(t) = ?$

Expected outcome: convolving with $\delta(t)$ leaves the signal unchanged – $\delta(t)$ is the identity element of convolution, just as 0 is the identity of addition.

$$x(t) * \delta(t) = \int_{-\infty}^{\infty} x(\tau) \delta(t - \tau) d\tau$$

By the sifting property, $\delta(t - \tau)$ picks out the integrand at the single point $\tau = t$ (the only place the impulse is nonzero), so

$$x(t) * \delta(t) = x(t).$$

More generally, $x(t) * \delta(t - t_0) = x(t - t_0)$: convolving with a shifted impulse just shifts the signal.

1.2 – Setting up: $h(t) = \delta(t + 2) + 2\delta(t + 1)$

$$y(t) = x(t) * h(t) = \int_{-\infty}^{\infty} x(\tau) [\delta(t - \tau + 2) + 2\delta(t - \tau + 1)] d\tau$$

Using the sifting property (each impulse in τ fires at $\tau = t + 2$ and $\tau = t + 1$):

$$y(t) = x(t + 2) + 2x(t + 1)$$

1.2 – Setting up: $h(t) = \delta(t + 2) + 2\delta(t + 1)$

$$y(t) = x(t) * h(t) = \int_{-\infty}^{\infty} x(\tau) [\delta(t - \tau + 2) + 2\delta(t - \tau + 1)] d\tau$$

Using the sifting property (each impulse in τ fires at $\tau = t + 2$ and $\tau = t + 1$):

$$y(t) = x(t + 2) + 2x(t + 1)$$

Recall

$$x(t) = \begin{cases} t + 1, & 0 \leq t \leq 1 \\ 2 - t, & 1 < t \leq 2 \\ 0, & \text{elsewhere} \end{cases}$$

Note x has two jump discontinuities: up from 0 to 1 at $t = 0$, and down from 2 to 1 at $t = 1$. These jumps will reappear (shifted) in $y(t)$.

1.2 – Building $x(t+2)$ and $x(t+1)$

Substitute $s = t + 2$ and $s = t + 1$ into the piecewise definition of $x(s)$:

$$x(t+2) = \begin{cases} t+3, & -2 \leq t \leq -1 \\ -t, & -1 < t \leq 0 \\ 0, & \text{elsewhere} \end{cases} \quad x(t+1) = \begin{cases} t+2, & -1 \leq t \leq 0 \\ 1-t, & 0 < t \leq 1 \\ 0, & \text{elsewhere} \end{cases}$$

1.2 – Building $x(t+2)$ and $x(t+1)$

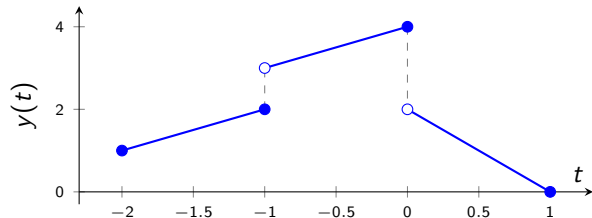
Substitute $s = t + 2$ and $s = t + 1$ into the piecewise definition of $x(s)$:

$$x(t+2) = \begin{cases} t+3, & -2 \leq t \leq -1 \\ -t, & -1 < t \leq 0 \\ 0, & \text{elsewhere} \end{cases} \quad x(t+1) = \begin{cases} t+2, & -1 \leq t \leq 0 \\ 1-t, & 0 < t \leq 1 \\ 0, & \text{elsewhere} \end{cases}$$

Adding $x(t+2) + 2x(t+1)$ region by region:

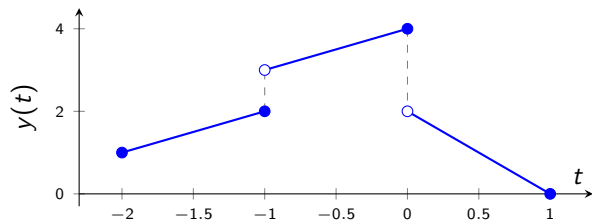
- $-2 \leq t < -1$: only $x(t+2)$ is active $\Rightarrow y = t + 3$.
- $-1 \leq t < 0$: both active $\Rightarrow y = (-t) + 2(t+2) = t + 4$.
- $0 \leq t \leq 1$: only $x(t+1)$ is active $\Rightarrow y = 2(1-t) = 2 - 2t$.

1.2 – Result: $y(t)$



$$y(t) = \begin{cases} t + 3, & -2 \leq t < -1 \\ t + 4, & -1 \leq t < 0 \\ 2 - 2t, & 0 \leq t \leq 1 \\ 0, & \text{elsewhere} \end{cases}$$

1.2 – Result: $y(t)$



$$y(t) = \begin{cases} t + 3, & -2 \leq t < -1 \\ t + 4, & -1 \leq t < 0 \\ 2 - 2t, & 0 \leq t \leq 1 \\ 0, & \text{elsewhere} \end{cases}$$

Checks: the jump at $t = -1$ (size +1) and at $t = 0$ (size -2) come from x 's own jumps, shifted and weighted by the impulse coefficients 1 and 2. Also, $\int y(t) dt = 6 = (\int x dt) (\int h) = 2 \cdot (1 + 2) \checkmark$

$$1.3 - x(t) = u(t - 3) - u(t - 5), \quad h(t) = e^{-3t}u(t)$$

(a) $y(t) = \int_3^5 h(t - \tau) d\tau$, and $h(t - \tau) \neq 0$ only for $\tau < t$. Three cases:

- $t < 3$: no overlap $\Rightarrow y(t) = 0$.
- $3 \leq t < 5$: integrate τ from 3 to t :

$$y(t) = \int_3^t e^{-3(t-\tau)} d\tau = \frac{1}{3} \left(1 - e^{-3(t-3)} \right)$$

- $t \geq 5$: integrate over the full pulse, τ from 3 to 5:

$$y(t) = \int_3^5 e^{-3(t-\tau)} d\tau = \frac{1}{3} \left(e^{-3(t-5)} - e^{-3(t-3)} \right)$$

Both pieces agree at $t = 5$: $\frac{1}{3}(1 - e^{-6}) \checkmark$

1.3 – (b), (c)

(b) $\frac{dx}{dt} = \delta(t - 3) - \delta(t - 5)$ (derivative of two steps), so

$$g(t) = \left(\frac{dx}{dt} \right) * h(t) = h(t - 3) - h(t - 5) = e^{-3(t-3)}u(t - 3) - e^{-3(t-5)}u(t - 5).$$

1.3 – (b), (c)

(b) $\frac{dx}{dt} = \delta(t-3) - \delta(t-5)$ (derivative of two steps), so

$$g(t) = \left(\frac{dx}{dt} \right) * h(t) = h(t-3) - h(t-5) = e^{-3(t-3)}u(t-3) - e^{-3(t-5)}u(t-5).$$

(c) Differentiate $y(t)$ from part (a) piece by piece:

- $3 < t < 5$: $\frac{dy}{dt} = e^{-3(t-3)}$, matching $g(t)$ there.
- $t > 5$: $\frac{dy}{dt} = e^{-3(t-3)} - e^{-3(t-5)}$, matching $g(t)$ there.

So $\boxed{g(t) = \frac{dy(t)}{dt}}$. This is a general LTI property: differentiation commutes with convolution,

$$\frac{d}{dt} [x(t) * h(t)] = \frac{dx}{dt} * h(t) = x(t) * \frac{dh}{dt}.$$

2.1 – Causal? Stable?

(a) $h[n] = \left(\frac{1}{5}\right)^n u[n]$

- Causal: **yes**, $h[n] = 0$ for $n < 0$.
- Stable: $\sum_{n=0}^{\infty} \left| \left(\frac{1}{5}\right)^n \right| = \frac{1}{1 - 1/5} = \frac{5}{4} < \infty \Rightarrow$ **stable**.

2.1 – Causal? Stable?

(a) $h[n] = \left(\frac{1}{5}\right)^n u[n]$

- Causal: **yes**, $h[n] = 0$ for $n < 0$.
- Stable: $\sum_{n=0}^{\infty} \left| \left(\frac{1}{5}\right)^n \right| = \frac{1}{1 - 1/5} = \frac{5}{4} < \infty \Rightarrow$ **stable**.

(b) $h(t) = te^{-t}u(t)$

- Causal: **yes**, $h(t) = 0$ for $t < 0$.
- Stable: $\int_0^{\infty} |t|e^{-t}dt = \Gamma(2) = 1 < \infty \Rightarrow$ **stable**.

2.2 – Block diagram: setting up

Two branches, $h_1(t) = u(t + 1)$ and $h_2(t) = u(t)$, are in **parallel** (their outputs are summed); the sum then feeds **in series** into $h_3(t) = e^{-4t}u(t - 2)$.

$$h(t) = [h_1(t) + h_2(t)] * h_3(t)$$

2.2 – Block diagram: setting up

Two branches, $h_1(t) = u(t+1)$ and $h_2(t) = u(t)$, are in **parallel** (their outputs are summed); the sum then feeds **in series** into $h_3(t) = e^{-4t}u(t-2)$.

$$h(t) = [h_1(t) + h_2(t)] * h_3(t)$$

First combine the parallel branches:

$$h_p(t) = u(t+1) + u(t) = \begin{cases} 0, & t < -1 \\ 1, & -1 \leq t < 0 \\ 2, & t \geq 0 \end{cases}$$

a two-step staircase that settles at the constant value 2.

2.2 – Convolution a step with $h_3(t)$

It is enough to convolve a single shifted step $u(t - a)$ with $h_3(t) = e^{-4t}u(t - 2)$, then add the two cases $a = -1$ and $a = 0$.

$$u(t - a) * h_3(t) = \int_a^\infty h_3(t - \tau) d\tau \stackrel{\sigma = t - \tau}{=} \int_{-\infty}^{t-a} h_3(\sigma) d\sigma$$

2.2 – Convolution a step with $h_3(t)$

It is enough to convolve a single shifted step $u(t - a)$ with $h_3(t) = e^{-4t}u(t - 2)$, then add the two cases $a = -1$ and $a = 0$.

$$u(t - a) * h_3(t) = \int_a^\infty h_3(t - \tau) d\tau \stackrel{\sigma = t - \tau}{=} \int_{-\infty}^{t-a} h_3(\sigma) d\sigma$$

Since $h_3(\sigma) \neq 0$ only for $\sigma \geq 2$, this is 0 unless $t - a \geq 2$, in which case

$$\int_2^{t-a} e^{-4\sigma} d\sigma = \frac{1}{4} \left(e^{-8} - e^{-4(t-a)} \right).$$

So define $\phi(t; a) = \frac{1}{4} \left(e^{-8} - e^{-4(t-a)} \right) u(t - a - 2)$. Then

$$h(t) = \phi(t; -1) + \phi(t; 0).$$

2.2 – Result

$$h(t) = \begin{cases} 0, & t < 1 \\ \frac{1}{4} \left(e^{-8} - e^{-4(t+1)} \right), & 1 \leq t < 2 \\ \frac{1}{4} \left(2e^{-8} - e^{-4(t+1)} - e^{-4t} \right), & t \geq 2 \end{cases}$$

(continuous at $t = 1$ and $t = 2$; check: both pieces give $\frac{1}{4}(e^{-8} - e^{-12})$ at $t = 2$.)

2.2 – Result

$$h(t) = \begin{cases} 0, & t < 1 \\ \frac{1}{4}(e^{-8} - e^{-4(t+1)}), & 1 \leq t < 2 \\ \frac{1}{4}(2e^{-8} - e^{-4(t+1)} - e^{-4t}), & t \geq 2 \end{cases}$$

(continuous at $t = 1$ and $t = 2$; check: both pieces give $\frac{1}{4}(e^{-8} - e^{-12})$ at $t = 2$.)

- **Causal? Yes:** $h(t) = 0$ for all $t < 0$ (in fact for $t < 1$).
- **Stable? No.** As $t \rightarrow \infty$, $h(t) \rightarrow \frac{1}{2}e^{-8}$, a *nonzero constant* rather than decaying to 0, so $\int |h(t)|dt$ diverges.

2.2 – Result

$$h(t) = \begin{cases} 0, & t < 1 \\ \frac{1}{4} \left(e^{-8} - e^{-4(t+1)} \right), & 1 \leq t < 2 \\ \frac{1}{4} \left(2e^{-8} - e^{-4(t+1)} - e^{-4t} \right), & t \geq 2 \end{cases}$$

(continuous at $t = 1$ and $t = 2$; check: both pieces give $\frac{1}{4}(e^{-8} - e^{-12})$ at $t = 2$.)

- **Causal? Yes:** $h(t) = 0$ for all $t < 0$ (in fact for $t < 1$).
- **Stable? No.** As $t \rightarrow \infty$, $h(t) \rightarrow \frac{1}{2}e^{-8}$, a *nonzero constant* rather than decaying to 0, so $\int |h(t)| dt$ diverges.

Why: h_1 and h_2 are each step responses – individually unstable, since a step never decays. Their parallel sum still carries this “DC” content. Cascading with the stable filter h_3 smooths it out but cannot remove it, so the overall impulse response settles to a nonzero constant instead of vanishing.

3.1 – Ideal filter: $H(j\omega) = \mathbb{K}\{|\omega| \leq \omega_0\}$

Conceptually: this is an **ideal (brick-wall) lowpass filter**. It passes every frequency component with $|\omega| \leq \omega_0$ through completely unchanged (gain 1, no phase shift), and completely removes every component with $|\omega| > \omega_0$.

- Sharp, discontinuous cutoff at $\omega = \omega_0$ – unlike physical filters.
- Purely real $\Rightarrow$ zero phase distortion inside the passband.
- Such an ideal brick-wall filter is not causal (its impulse response is a sinc, nonzero for $t < 0$) – a useful idealization, but not physically realizable exactly.

3.2 – Frequency response of $h(t) = e^{-t}u(t)$

$$H(j\omega) = \int_{-\infty}^{\infty} e^{-j\omega\tau} h(\tau) d\tau = \int_0^{\infty} e^{-(1+j\omega)\tau} d\tau = \frac{1}{1+j\omega}$$

(converges since $\text{Re}\{1+j\omega\} = 1 > 0$).

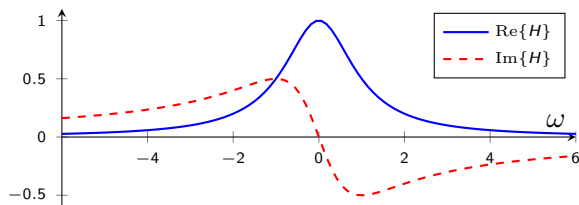
3.2 – Frequency response of $h(t) = e^{-t}u(t)$

$$H(j\omega) = \int_{-\infty}^{\infty} e^{-j\omega\tau} h(\tau) d\tau = \int_0^{\infty} e^{-(1+j\omega)\tau} d\tau = \frac{1}{1+j\omega}$$

(converges since $\text{Re}\{1+j\omega\} = 1 > 0$). Rationalize: $H(j\omega) = \frac{1-j\omega}{1+\omega^2}$, so

$$\text{Re}\{H(j\omega)\} = \frac{1}{1+\omega^2}, \quad \text{Im}\{H(j\omega)\} = \frac{-\omega}{1+\omega^2}.$$

3.2(a) – Plot and comparison with the ideal filter



- $\text{Re}\{H\}$ is even, peaks at 1 ($\omega = 0$), decays smoothly – a gradual lowpass roll-off, not a brick wall.
- $\text{Im}\{H\}$ is odd, zero at $\omega = 0$, extrema $\pm \frac{1}{2}$ at $\omega = \mp 1$: a nonzero **phase shift** at every nonzero frequency.
- Unlike 3.1's filter, this one is causal and realizable (a simple RC-type lowpass), at the cost of a gentle rather than sharp cutoff and added phase distortion.

3.2(b) – Output for $x(t) = e^{3jt}$: setting up

Here $\omega = 3$: $H(j3) = \frac{1}{1+3j} = \frac{1-3j}{10} = 0.1 - 0.3j$.

$$|H(j3)| = \frac{1}{\sqrt{10}} \approx 0.316, \quad \angle H(j3) = -\arctan(3) \approx -71.6^\circ = -1.249 \text{ rad}$$

3.2(b) – Output for $x(t) = e^{3jt}$: setting up

Here $\omega = 3$: $H(j3) = \frac{1}{1+3j} = \frac{1-3j}{10} = 0.1 - 0.3j$.

$$|H(j3)| = \frac{1}{\sqrt{10}} \approx 0.316, \quad \angle H(j3) = -\arctan(3) \approx -71.6^\circ = -1.249 \text{ rad}$$

Write $H(j3)$ in magnitude-phase (polar) form:

$$H(j3) = |H(j3)| e^{j\angle H(j3)} = \frac{1}{\sqrt{10}} e^{-j\arctan 3}$$

$$y(t) = H(j3) e^{3jt} = \frac{1}{\sqrt{10}} e^{-j\arctan 3} e^{3jt} = \frac{1}{\sqrt{10}} e^{j(3t - \arctan 3)}$$

Let $\varphi = \arctan 3$ for short. We still need to split this into real and imaginary parts – that's where the angle-difference formulas come in.

3.2(b) – Expanding with the angle-difference formulas

Euler's formula gives $y(t) = \frac{1}{\sqrt{10}} [\cos(3t - \varphi) + j \sin(3t - \varphi)]$. Expand each using

$$\cos(3t - \varphi) = \cos(3t) \cos \varphi + \sin(3t) \sin \varphi, \quad \sin(3t - \varphi) = \sin(3t) \cos \varphi - \cos(3t) \sin \varphi.$$

3.2(b) – Expanding with the angle-difference formulas

Euler's formula gives $y(t) = \frac{1}{\sqrt{10}} [\cos(3t - \varphi) + j \sin(3t - \varphi)]$. Expand each using

$$\cos(3t - \varphi) = \cos(3t) \cos \varphi + \sin(3t) \sin \varphi, \quad \sin(3t - \varphi) = \sin(3t) \cos \varphi - \cos(3t) \sin \varphi.$$

Since $\varphi = \arctan 3$ is the angle of the point $(1, 3)$ (adjacent = 1, opposite = 3, hypotenuse = $\sqrt{10}$, from rationalizing $H(j3)$ above):

$$\cos \varphi = \frac{1}{\sqrt{10}}, \quad \sin \varphi = \frac{3}{\sqrt{10}}.$$

3.2(b) – Expanding with the angle-difference formulas

Euler's formula gives $y(t) = \frac{1}{\sqrt{10}} [\cos(3t - \varphi) + j \sin(3t - \varphi)]$. Expand each using

$$\cos(3t - \varphi) = \cos(3t) \cos \varphi + \sin(3t) \sin \varphi, \quad \sin(3t - \varphi) = \sin(3t) \cos \varphi - \cos(3t) \sin \varphi.$$

Since $\varphi = \arctan 3$ is the angle of the point $(1, 3)$ (adjacent = 1, opposite = 3, hypotenuse = $\sqrt{10}$, from rationalizing $H(j3)$ above):

$$\cos \varphi = \frac{1}{\sqrt{10}}, \quad \sin \varphi = \frac{3}{\sqrt{10}}.$$

Substituting into the real part:

$$\operatorname{Re}\{y(t)\} = \frac{1}{\sqrt{10}} \left[\cos(3t) \cdot \frac{1}{\sqrt{10}} + \sin(3t) \cdot \frac{3}{\sqrt{10}} \right] = \frac{1}{10} \cos(3t) + \frac{3}{10} \sin(3t)$$

and into the imaginary part:

$$\operatorname{Im}\{y(t)\} = \frac{1}{\sqrt{10}} \left[\sin(3t) \cdot \frac{1}{\sqrt{10}} - \cos(3t) \cdot \frac{3}{\sqrt{10}} \right] = \frac{1}{10} \sin(3t) - \frac{3}{10} \cos(3t)$$

3.2(b) – Expanding with the angle-difference formulas

Euler's formula gives $y(t) = \frac{1}{\sqrt{10}} [\cos(3t - \varphi) + j \sin(3t - \varphi)]$. Expand each using

$$\cos(3t - \varphi) = \cos(3t) \cos \varphi + \sin(3t) \sin \varphi, \quad \sin(3t - \varphi) = \sin(3t) \cos \varphi - \cos(3t) \sin \varphi.$$

Since $\varphi = \arctan 3$ is the angle of the point (1, 3) (adjacent = 1, opposite = 3, hypotenuse = $\sqrt{10}$, from rationalizing $H(j3)$ above):

$$\cos \varphi = \frac{1}{\sqrt{10}}, \quad \sin \varphi = \frac{3}{\sqrt{10}}.$$

Substituting into the real part:

$$\operatorname{Re}\{y(t)\} = \frac{1}{\sqrt{10}} \left[\cos(3t) \cdot \frac{1}{\sqrt{10}} + \sin(3t) \cdot \frac{3}{\sqrt{10}} \right] = \frac{1}{10} \cos(3t) + \frac{3}{10} \sin(3t)$$

and into the imaginary part:

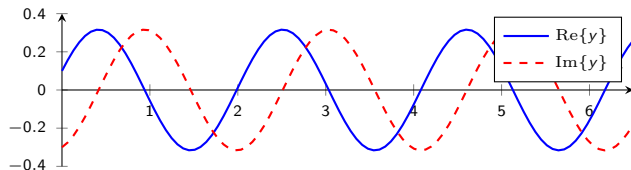
$$\operatorname{Im}\{y(t)\} = \frac{1}{\sqrt{10}} \left[\sin(3t) \cdot \frac{1}{\sqrt{10}} - \cos(3t) \cdot \frac{3}{\sqrt{10}} \right] = \frac{1}{10} \sin(3t) - \frac{3}{10} \cos(3t)$$

Cross-check (direct expansion, no phase form):

$$y(t) = (0.1 - 0.3j)(\cos 3t + j \sin 3t) = [0.1 \cos 3t + 0.3 \sin 3t] + j[0.1 \sin 3t - 0.3 \cos 3t] - \text{matches exactly.}$$

3.2(b) – Result and plot

$$\operatorname{Re}\{y(t)\} = 0.1 \cos(3t) + 0.3 \sin(3t), \quad \operatorname{Im}\{y(t)\} = 0.1 \sin(3t) - 0.3 \cos(3t)$$



Both are sinusoids at the same frequency $\omega = 3$, scaled by $|H(j3)|$ and shifted by its phase.

3.2(c) – Output for $x(t) = \sin(4t) + \frac{1}{2} \cos(2t)$: setting up

Decompose into complex exponentials:

$$\sin(4t) = \frac{e^{j4t} - e^{-j4t}}{2j}, \quad \frac{1}{2} \cos(2t) = \frac{e^{j2t} + e^{-j2t}}{4}$$

3.2(c) – Output for $x(t) = \sin(4t) + \frac{1}{2} \cos(2t)$: setting up

Decompose into complex exponentials:

$$\sin(4t) = \frac{e^{j4t} - e^{-j4t}}{2j}, \quad \frac{1}{2} \cos(2t) = \frac{e^{j2t} + e^{-j2t}}{4}$$

Now $H(j\omega)$:

$$y(t) = \underbrace{\frac{1}{2j} [H(j4)e^{j4t} - H(-j4)e^{-j4t}]}_{\text{from } \sin(4t)} + \underbrace{\frac{1}{4} [H(j2)e^{j2t} + H(-j2)e^{-j2t}]}_{\text{from } \frac{1}{2} \cos(2t)}$$

3.2(c) – Output for $x(t) = \sin(4t) + \frac{1}{2} \cos(2t)$: setting up

Decompose into complex exponentials:

$$\sin(4t) = \frac{e^{j4t} - e^{-j4t}}{2j}, \quad \frac{1}{2} \cos(2t) = \frac{e^{j2t} + e^{-j2t}}{4}$$

Now $H(j\omega)$:

$$y(t) = \underbrace{\frac{1}{2j} [H(j4)e^{j4t} - H(-j4)e^{-j4t}]}_{\text{from } \sin(4t)} + \underbrace{\frac{1}{4} [H(j2)e^{j2t} + H(-j2)e^{-j2t}]}_{\text{from } \frac{1}{2} \cos(2t)}$$

We need $H(j4)$, $H(j2)$, and their conjugates $H(-j4)$, $H(-j2)$. Since $h(t)$ is real, $H(-j\omega) = H(j\omega)^*$ (conjugate symmetry) – so we only need to compute $H(j4)$ and $H(j2)$.

3.2(c) – Computing $H(j4)$ and $H(j2)$

Using $H(j\omega) = \frac{1}{1+j\omega}$ from before:

$$H(j4) = \frac{1}{1+4j} = \frac{1-4j}{17}, \quad H(j2) = \frac{1}{1+2j} = \frac{1-2j}{5}$$

3.2(c) – Computing $H(j4)$ and $H(j2)$

Using $H(j\omega) = \frac{1}{1+j\omega}$ from before:

$$H(j4) = \frac{1}{1+4j} = \frac{1-4j}{17}, \quad H(j2) = \frac{1}{1+2j} = \frac{1-2j}{5}$$

In polar form, with $\varphi_4 = \arctan 4$ and $\varphi_2 = \arctan 2$:

$$H(j4) = \frac{1}{\sqrt{17}} e^{-j\varphi_4}, \quad H(j2) = \frac{1}{\sqrt{5}} e^{-j\varphi_2}$$

reading off $\cos \varphi_4 = \frac{1}{\sqrt{17}}$, $\sin \varphi_4 = \frac{4}{\sqrt{17}}$ and $\cos \varphi_2 = \frac{1}{\sqrt{5}}$, $\sin \varphi_2 = \frac{2}{\sqrt{5}}$ (same triangle trick as in 3.2(b)).

3.2(c) – Computing $H(j4)$ and $H(j2)$

Using $H(j\omega) = \frac{1}{1+j\omega}$ from before:

$$H(j4) = \frac{1}{1+4j} = \frac{1-4j}{17}, \quad H(j2) = \frac{1}{1+2j} = \frac{1-2j}{5}$$

In polar form, with $\varphi_4 = \arctan 4$ and $\varphi_2 = \arctan 2$:

$$H(j4) = \frac{1}{\sqrt{17}} e^{-j\varphi_4}, \quad H(j2) = \frac{1}{\sqrt{5}} e^{-j\varphi_2}$$

reading off $\cos \varphi_4 = \frac{1}{\sqrt{17}}$, $\sin \varphi_4 = \frac{4}{\sqrt{17}}$ and $\cos \varphi_2 = \frac{1}{\sqrt{5}}$, $\sin \varphi_2 = \frac{2}{\sqrt{5}}$ (same triangle trick as in 3.2(b)). By conjugate symmetry, $H(-j4) = H(j4)^* = \frac{1}{\sqrt{17}} e^{+j\varphi_4}$ and $H(-j2) = H(j2)^* = \frac{1}{\sqrt{5}} e^{+j\varphi_2}$.

3.2(c) – Collapsing each conjugate pair

For any complex $A = |A|e^{j\angle A}$, recall $z - z^* = 2j \operatorname{Im}\{z\}$ and $z + z^* = 2 \operatorname{Re}\{z\}$ with $z = Ae^{j\theta}$. So:

$$\frac{1}{2j} \left[H(j4)e^{j4t} - H(j4)^*e^{-j4t} \right] = \operatorname{Im}\{H(j4)e^{j4t}\}$$

$$\frac{1}{4} \left[H(j2)e^{j2t} + H(j2)^*e^{-j2t} \right] = \frac{1}{2} \operatorname{Re}\{H(j2)e^{j2t}\}$$

3.2(c) – Collapsing each conjugate pair

For any complex $A = |A|e^{j\angle A}$, recall $z - z^* = 2j \operatorname{Im}\{z\}$ and $z + z^* = 2 \operatorname{Re}\{z\}$ with $z = Ae^{j\theta}$. So:

$$\frac{1}{2j} \left[H(j4)e^{j4t} - H(j4)^*e^{-j4t} \right] = \operatorname{Im}\{H(j4)e^{j4t}\}$$

$$\frac{1}{4} \left[H(j2)e^{j2t} + H(j2)^*e^{-j2t} \right] = \frac{1}{2} \operatorname{Re}\{H(j2)e^{j2t}\}$$

Substituting the polar forms:

$$\operatorname{Im}\{H(j4)e^{j4t}\} = \operatorname{Im}\left\{ \frac{1}{\sqrt{17}} e^{j(4t-\varphi_4)} \right\} = \frac{1}{\sqrt{17}} \sin(4t - \varphi_4)$$

$$\frac{1}{2} \operatorname{Re}\{H(j2)e^{j2t}\} = \frac{1}{2} \operatorname{Re}\left\{ \frac{1}{\sqrt{5}} e^{j(2t-\varphi_2)} \right\} = \frac{1}{2\sqrt{5}} \cos(2t - \varphi_2)$$

$$\text{So } y(t) = \frac{1}{\sqrt{17}} \sin(4t - \varphi_4) + \frac{1}{2\sqrt{5}} \cos(2t - \varphi_2)$$

3.2(c) – Expanding with the angle-difference formulas

Exactly as in 3.2(b), expand each shifted sinusoid:

$$\sin(4t - \varphi_4) = \sin(4t) \cos \varphi_4 - \cos(4t) \sin \varphi_4, \quad \cos(2t - \varphi_2) = \cos(2t) \cos \varphi_2 + \sin(2t) \sin \varphi_2$$

3.2(c) – Expanding with the angle-difference formulas

Exactly as in 3.2(b), expand each shifted sinusoid:

$$\sin(4t - \varphi_4) = \sin(4t) \cos \varphi_4 - \cos(4t) \sin \varphi_4, \quad \cos(2t - \varphi_2) = \cos(2t) \cos \varphi_2 + \sin(2t) \sin \varphi_2$$

Substitute $\cos \varphi_4 = \frac{1}{\sqrt{17}}$, $\sin \varphi_4 = \frac{4}{\sqrt{17}}$, $\cos \varphi_2 = \frac{1}{\sqrt{5}}$, $\sin \varphi_2 = \frac{2}{\sqrt{5}}$:

$$\frac{1}{\sqrt{17}} \sin(4t - \varphi_4) = \frac{1}{\sqrt{17}} \left[\sin(4t) \cdot \frac{1}{\sqrt{17}} - \cos(4t) \cdot \frac{4}{\sqrt{17}} \right] = \frac{1}{17} \sin(4t) - \frac{4}{17} \cos(4t)$$

$$\frac{1}{2\sqrt{5}} \cos(2t - \varphi_2) = \frac{1}{2\sqrt{5}} \left[\cos(2t) \cdot \frac{1}{\sqrt{5}} + \sin(2t) \cdot \frac{2}{\sqrt{5}} \right] = \frac{1}{10} \cos(2t) + \frac{1}{5} \sin(2t)$$

3.2(c) – Expanding with the angle-difference formulas

Exactly as in 3.2(b), expand each shifted sinusoid:

$$\sin(4t - \varphi_4) = \sin(4t) \cos \varphi_4 - \cos(4t) \sin \varphi_4, \quad \cos(2t - \varphi_2) = \cos(2t) \cos \varphi_2 + \sin(2t) \sin \varphi_2$$

Substitute $\cos \varphi_4 = \frac{1}{\sqrt{17}}$, $\sin \varphi_4 = \frac{4}{\sqrt{17}}$, $\cos \varphi_2 = \frac{1}{\sqrt{5}}$, $\sin \varphi_2 = \frac{2}{\sqrt{5}}$:

$$\frac{1}{\sqrt{17}} \sin(4t - \varphi_4) = \frac{1}{\sqrt{17}} \left[\sin(4t) \cdot \frac{1}{\sqrt{17}} - \cos(4t) \cdot \frac{4}{\sqrt{17}} \right] = \frac{1}{17} \sin(4t) - \frac{4}{17} \cos(4t)$$

$$\frac{1}{2\sqrt{5}} \cos(2t - \varphi_2) = \frac{1}{2\sqrt{5}} \left[\cos(2t) \cdot \frac{1}{\sqrt{5}} + \sin(2t) \cdot \frac{2}{\sqrt{5}} \right] = \frac{1}{10} \cos(2t) + \frac{1}{5} \sin(2t)$$

$$y(t) = \frac{1}{17} \sin(4t) - \frac{4}{17} \cos(4t) + \frac{1}{10} \cos(2t) + \frac{1}{5} \sin(2t)$$

Numeric check: $\frac{1}{17} = 0.0588$, $\frac{4}{17} = 0.2353$, $\frac{1}{10} = 0.1$, $\frac{1}{5} = 0.2$ – expanding $0.243 \sin(4t - 1.326) + 0.224 \cos(2t - 1.107)$ the same way reproduces these same four coefficients.

3.2(c) – Result

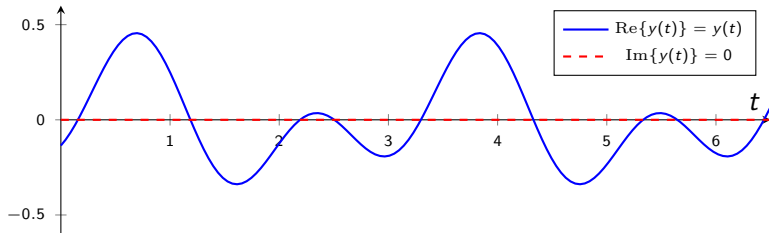
$$y(t) = 0.243 \sin(4t - 1.326) + 0.224 \cos(2t - 1.107) = \frac{1}{17} \sin(4t) - \frac{4}{17} \cos(4t) + \frac{1}{10} \cos(2t) + \frac{1}{5} \sin(2t)$$

3.2(c) – Result

$$y(t) = 0.243 \sin(4t - 1.326) + 0.224 \cos(2t - 1.107) = \frac{1}{17} \sin(4t) - \frac{4}{17} \cos(4t) + \frac{1}{10} \cos(2t) + \frac{1}{5} \sin(2t)$$

Real and imaginary parts: because $x(t)$ is real and $h(t)$ is real, $y(t)$ is *purely real* – $\text{Re}\{y(t)\} = y(t)$ as above and $\text{Im}\{y(t)\} = 0$. This is different from 3.2(b), where the input itself was complex. Notice also this fell out automatically: every term above collapsed into a real sin or cos, with no leftover j 's, precisely because we paired up each exponential with its conjugate.

3.2(c) – Plot



The imaginary part is identically the zero line for all t – there is nothing frequency-dependent to plot, since a real input into a real system can only ever produce a real output. Contrast this flat $\text{Im}\{y\}$ with the nonzero sinusoid seen in 3.2(b), where the input e^{3jt} was itself complex.